MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER,
EXCLUSIVE-OR GATES, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/26(USAF), dated 15 August 1972.

PAGE 1

* 1.2, delete (including example of part number) and substitute:

"1.2 Part number. The complete part number shall be in accordance with MIL-M-38510."

PAGE 3

TABLE I, I_{IH2} test, maximum limits column: Delete "80" and substitute "100".

TABLE I, I_{IL} test, minimum limits column: Add "-120"; maximum limits column: delete "-0.36" and substitute "-360"; units column: delete "mA" and substitute "µA".

TABLE I, I_{OS} and I_{CCH} tests, conditions column: Delete "2/".

TABLE I, t_{PHL1} test, maximum limits column: Delete "165" and substitute "275".

TABLE I, t_{PHL2} test, maximum limits column: Delete "80" and substitute "110".

TABLE I, t_{PLH1} test, maximum limits column: Delete "180" and substitute "250".

TABLE I, t_{PLH2} test, maximum limits column: Delete "110" and substitute "150".

PAGE 4

* TABLE II, test requirement column: Delete "Groups B and C end point electrical parameters (Method 5005)" and substitute "Groups C and D end point electrical parameters (Method 5005)".

* 4.2, second sentence: Delete "A, B, and C" and substitute "A, B, C, and D". Delete third sentence.

* 4.3: Delete subparagraphs (a), (b), (c), (e), and (g).

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PAGE 5

* 4.4.1, first sentence: Delete "consist of the test subgroups and LTPD values shown in" and substitute "be in accordance with".

* 4.4.2, delete and substitute:

"Group B inspection. Group B inspection shall be in accordance with table II of Method 5005 of MIL-STD-883."

* 4.4.3: Delete title "Group C inspection" and substitute "Groups C and D inspections". Delete first sentence and substitute "Groups C and D inspections shall be in accordance with tables III and IV of Method 5005 of MIL-STD-883 and as follows:" Delete subparagraphs (c), (d), and (f). Subparagraph (b): Delete "Subgroups 7 and 8" and substitute "Subgroups 3 and 4".

PAGE 6

FIGURE 1, logic diagram and terminal connections: Delete and substitute the following:

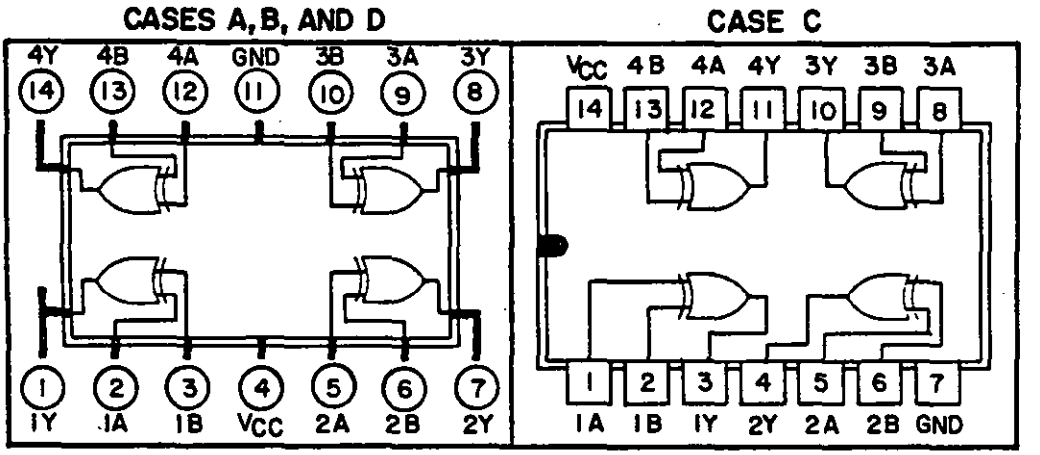
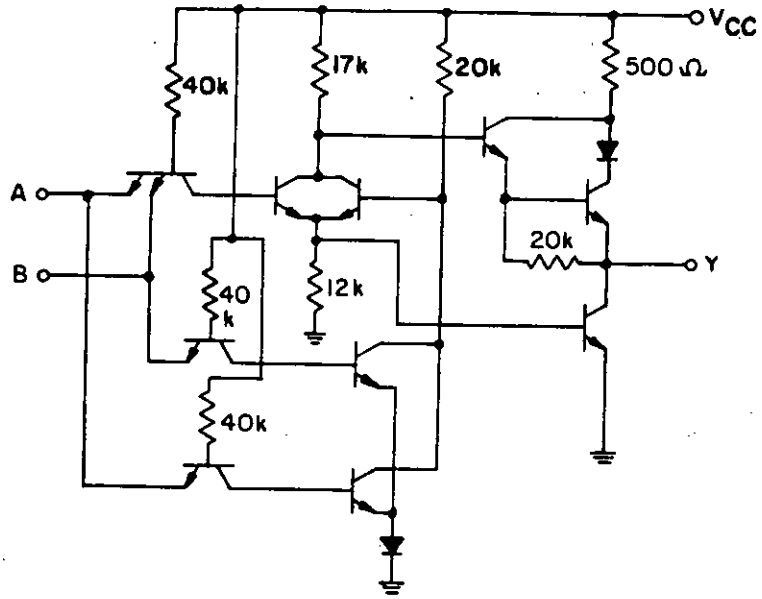


*FIGURE 3, schematic circuit: Add "circuit A" after schematic circuit; add new circuit (B) as shown on page 3 of this amendment.

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NOTE: All resistance values are nominal.

FIGURE 3. Schematic circuit - circuit B.

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FIGURE 5, switching time test circuit: Delete and substitute figure 5 as shown on page 4 of this amendment.

PAGE 10

TABLE III, tests 31 and 32, column 3A: Add "GND".

* TABLE III, tests 25 thru 32, maximum limits column: Delete "80" and substitute "100".

* TABLE III, tests 33 thru 40, maximum limits column: Delete ".36 mA" and substitute "-360 µA".

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TABLE III, tests 48 thru 51, maximum limits column: Delete "105" and substitute "165".

TABLE III, tests 52 thru 55, maximum limits column: Delete "50" and substitute "66".

TABLE III, tests 56 thru 59, maximum limits column: Delete "115" and substitute "150".

TABLE III, tests 60 thru 63, maximum limits column: Delete "70" and substitute "90".

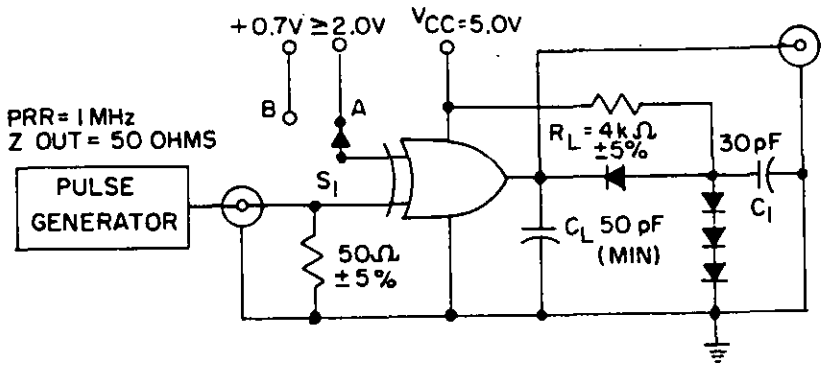
TABLE III, tests 64 thru 67, maximum limits column: Delete "165" and substitute "275".

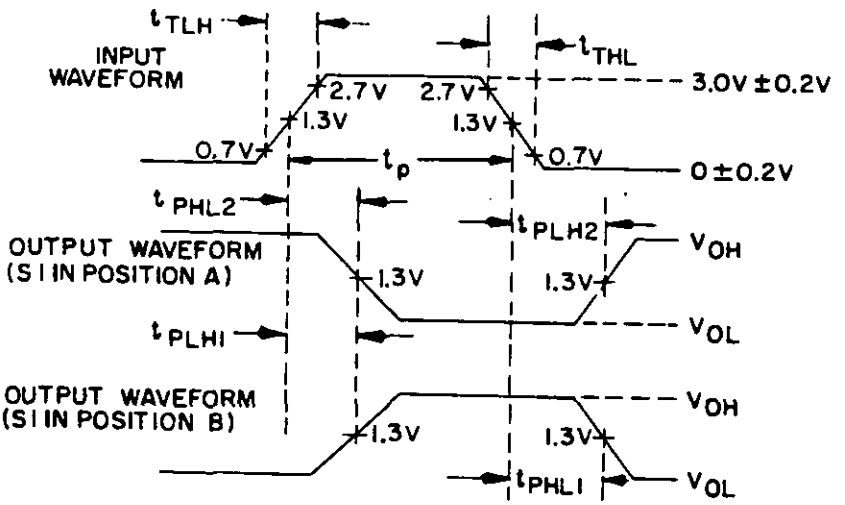
TABLE III, tests 68 thru 71, maximum limits column: Delete "80" and substitute "110".

TABLE III, tests 72 thru 75, maximum limits column: Delete "180" and substitute "250".

TABLE III, tests 76 thru 79, maximum limits column: Delete "110" and substitute "150".

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VOLTAGE WAVEFORMS

NOTES:
1. The generator has the following characteristics: V_{gen} = 3 V, $t_{TLH} = t_{THL} \leq 25$ ns, t_p = 1.0 µs, PRR ≤ 500 kHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064, or equivalent.
3. C_L includes probe and jig capacitance.
4. Each gate tested separately.

FIGURE 5. Switching time test circuit.

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6.5, delete and substitute:

"6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits procured to Government logistic support will be procured to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 17, 85
DSA - ES

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

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